Submission details

Grade: A

TEST 1 - Week 2

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Attempts 1

Allowed attempts 2

Status	Finished
Started	Wednesday, 30 September 2026, 10:32 AM
Completed	Wednesday, 30 September 2026, 7:47 PM
Duration	9 hours 14 mins
Marks	5.00/5.00
Grade	**15.00** out of 15.00 (**100**%)

Question 1

Correct

Mark 1.00 out of 1.00

Prove by induction that for all $n \in \mathbb{N}$ the following property holds:

$$\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\ldots+\frac{1}{(2n-1)(2n+1)}=\frac{n}{2n+1}$$

In the induction step, we proceed as follows:

- assuming that the induction hypothesis holds $P(n)$
- we have to prove the induction thesis, that is, we have to prove that $P(n+1)$ holds

Find which of the following statements corresponds to the induction thesis.

- i. $\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\ldots+\frac{1}{(2n+1)(2n+3)}=\frac{n+1}{2n+1}$
- ii. $\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\ldots+\frac{1}{(2n-1)(2n+1)}=\frac{n}{2n+1}$
- iii. $\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\ldots+\frac{1}{(2n)(2n+2)}=\frac{n+1}{2n+3}$
- iv. $\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\ldots+\frac{1}{(2n+1)(2n+3)}=\frac{n+1}{2n+3}$ ✓

Respuesta correcta

The induction thesis, $P(n+1)$, is obtained substituting n by $n+1$ in the given expression. Therefo

$P(n+1)$: $\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\ldots+\frac{1}{(2n+1)(2n+3)}=\frac{n+1}{2n+3}$

The correct answer is: $\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\ldots+\frac{1}{(2n+1)(2n+3)}=\frac{n+1}{2n+3}$

Question 2

Correct

Mark 1.00 out of 1.00

Perform the following operation with complex numbers in standard form:

$$z = \frac{(-4-4\cdot i)-(5-4\cdot i)\cdot(3-3\cdot i)}{1-2\cdot i}$$

Write it down in the form $a+bi$, where a and b are the real and imaginary parts of z.

$z =$ -53/5 ✓ + 9/5 ✓ i

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